

Technical report: phone captures to dimensioned, stitched floor plans

Final code 525a70a, 5 Oct 2026. The task: turn what a phone captures (a few photos per room, a walkthrough video, or a LiDAR scan) into one stitched floor plan of the property, with every wall, opening and ceiling height measured to the centimetre and an honest 95% interval on each number. This report explains each method we use, why it helps that task, and what it measurably changes. Fresh numbers come from `results/summary.json` (the benchmark, `BENCHMARK_REPORT.md`); method numbers cite their decision entry (D-xxx in `DECISIONS.md`). Paths are relative to `docs/`.

1. Architecture

Three front-ends, one back-end (D-003). Each tier only has to produce a metric 3D scene; everything after the scene is shared, so a fix to plan extraction helps every tier, and every tier returns the same `plan.json`.

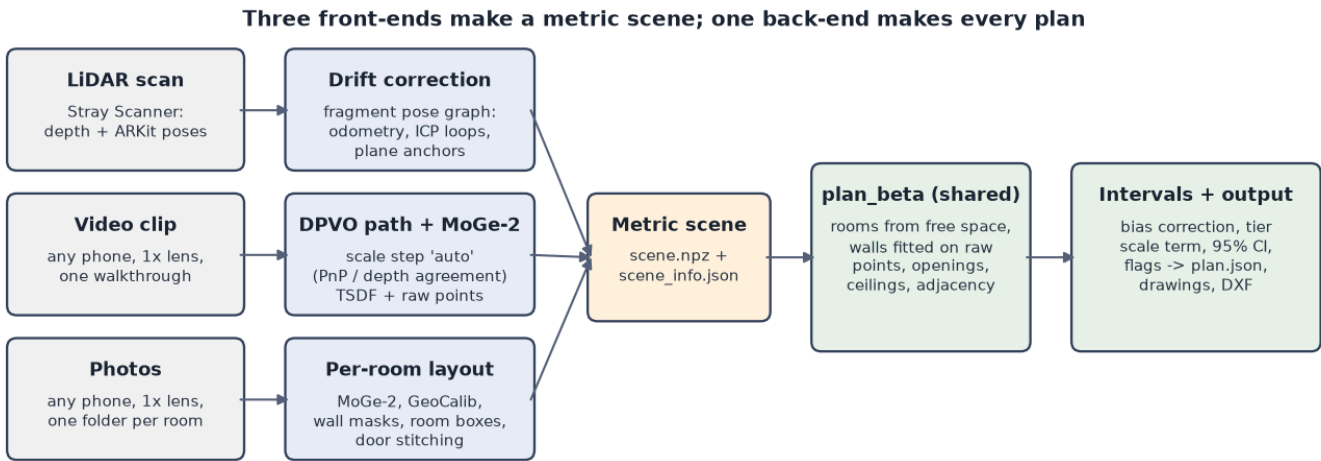


Figure 1. Architecture. Each tier only has to deliver a metric scene; the plan, the intervals and the drawings are shared. One command runs any tier: `python scripts/run_capture.py <capture> --tier lidar|video|photo --out <dir>`.

2. Tiers and the device matrix

The tiers differ in where metric scale comes from, and so in how wide their intervals must be (`DEVICE_MATRIX.md`).

Tier	Hardware	Metric source	Measured accuracy (fresh runs)
LiDAR	iPhone 12 Pro+ / iPad Pro 2020+ (Stray Scanner)	LiDAR depth, ARKit poses + our drift correction	k65: median wall error 1.8%, ceilings +1.1 to +1.4 cm, IoU 0.93; laser rooms 8/8 within 1.5 cm (cited)
Photo	any phone, 1x lens, 6–8 photos per room	MoGe-2 depth per photo, GeoCalib, EXIF focal	own bedroom (dim) 5/6 walls within 8%, footprint +0.6%; k65 stitched footprint –3.8%
Video	any phone, 1x lens	DPVO path, MoGe-2 depth, scale step "auto"	own flat: footprint –2.3% to +11.9%, walls 0–3 of 14 within 3%

3. The methods, and what each one buys

3.1 Shared back end

Method	What it is and how it works	Why it helps this task	What it changes
Plan extractor v2 (<code>plan_beta</code>)	Space-first: free floor space becomes rooms (watershed), split along fitted wall lines; walls, openings, ceilings and adjacency are then measured per room	It never invents space: a room exists only where the camera saw floor	Chosen over a wall-first extractor that added 5–9 m² of rooms nobody entered (D-010, D-018); wall lines as cuts took matched rooms from 6 to 8/8 (<code>modules/plan_beta.md</code> v2.4)
Raw-point wall fitting	Each wall is a line fitted to the raw depth points behind it, weighted by $1/\sigma^2$ of the sensor noise, not to a mesh or raster	Centimetre walls need the measurements themselves; a raster quantises them	Plane-fit error 0.04–0.08 mm: the fit is never the limit (<code>modules/arkitscenes_validation.md</code> §6)
remove_jogs loop fix	The step that snaps small jogs out of an outline stops when a snap returns the same corners (capped)	A plan must always finish; a stuck step gave no plan	Stitched k38 plan steps: over the 300 s limit → 2.7–2.9 s; the 16 plans that finished before are identical (D-088 follow-up, 23324d9)

Method	What it is and how it works	Why it helps this task	What it changes
95% intervals	Every number carries fit $\oplus$ sensor $\oplus$ drift $\oplus$ tier scale term; video joins without verified geometry get a 20% σ floor and reliability = low (D-016, D-034)	The brief wants honest intervals; a camera tier must say it is less sure	Photo intervals hold the truth 1.00 / 1.00 / 0.97 of the time (own dim, own lit, k65); LiDAR 0.70 and video 0.32–0.76 do not (§5)

3.2 LiDAR tier

Pose-graph drift correction (D-019, `modules/drift.md`). *What*: ARKit's poses drift (up to 5.95° of heading on floor_only), so a wall seen twice comes out twice. *How*: the walk is cut into 1.5 m / 5 s fragments; a pose graph joins them with ARKit odometry, ICP loop closures checked on fitness, RMSE and degeneracy, and plane anchors (each fragment's wall direction and floor level); it is solved robustly (IRLS-Cauchy) in 4 DoF, because gravity is observed. ICP runs single-threaded so the same input gives the same plan (D-029). *Why*: stitching a whole flat from one walk is exactly where drift shows. *Change* (ablation, fresh):

Capture	Footprint on / off	Rooms on / off	Cited: revisit surface gap on / off
floor_only	61.9 / 52.6 m ²	8 / 7	2.27 / 14.29 cm
with_ceiling	62.5 / 61.9 m ²	8 / 9	1.15 / 2.83 cm

Without it floor_only loses a room and 15% of its area. The ceiling never enters the solve, so it is an independent witness: its spread drops from 6.04 to 0.54 cm on with_ceiling (`modules/benchmark_lidar_sample.md` §3).

Inward-bias correction (D-021, D-027). *What*: Apple's LiDAR puts surfaces 0.68 cm into the room, so every room measures about 1.4 cm short. *How*: each wall face moves out by the bias, corner-aware: at a reflex corner moving out shortens a wall, so the shift follows the corner type. *Why*: at 1.5 cm gates a 1.4 cm systematic error is the whole budget. *Change*: pre-registered on hold-out rooms, the mean error went from −1.25 to +0.11 cm; the corner-aware version took laser walls within 1.5 cm from 82% to 94% (`modules/benchmark_arkitscenes_accuracy.md` §4.3). Fresh k65 ceilings: +1.1 to +1.4 cm, all five inside 1.5 cm.

3.3 Photo tier

The photo tier measures each room from one spot (a turning series of stills) and joins rooms at doors. A bad photo can only spoil its own room, which is why it works on the own flat where video does not (D-080).

Method	How it works	Why it helps	What it changes
Metric depth per photo (MoGe-2)	A learned model gives metric depth and a point map from a single image	Scale without LiDAR and without a reference object (D-067)	Sizes inside one frame of the own video within −5.8% to +0.9% of tape (FIX_LOOP.md, rejected alternatives)
Gravity (GeoCalib)	Estimates each photo's up direction and field of view	Walls become vertical planes; tilt tells a ceiling photo from a turning one	Needed by every later step; no separate ablation
Matching and pose graph (ALIKED + LightGlue)	Learned features link photos of one spin and doorway pairs (taken within seconds, found from EXIF times)	Places photos without depth or poses from the phone	Doorway pairs 9/9 found, 0 false (<code>modules/photo_tier.md</code> v2.5)
Per-room boxes from wall points	Each room is fitted as a Manhattan box from its own spin's wall points	A spin is a small panorama: no match across rooms is needed	Base of every room; refined by the rules below
Wall masks (SegFormer-B5, ADE20K)	Only pixels the segmenter calls "wall" define a side	Furniture in front of a wall stops being the wall	k65 footprint −14.9% → −8.3%, k38 −8.0% → −3.8% (D-060)
Focal from a sensor table	EXIF 35 mm focal with the standard diagonal; for the OnePlus Nord, which writes none, the sensor's 8.0 mm diagonal (<code>photo/images.py</code>)	Wrong focal = wrong scale	Without a focal the scale was 22% off (D-022); the Nord writes no 35 mm focal, so the table supplies it (D-048)
Ceiling-photo rule (D-074)	The last photo of a series tilted up > 10° is the ceiling photo (18° elsewhere)	My phone's ceiling photos were 16.6° up and were taken as turning photos	Own lit: walls median 44.0% → 9.1%, ceiling 2.314 → 2.514 m
Far-wall rule (D-077)	A farther wall peak with ≥ 0.8 m of wall replaces a nearer surface (wardrobe, door leaf); sides only move out	Bedrooms have wardrobes against walls	Own area −17% → −4% (lit), −31% → −4% (dim); k65 unchanged
Room shapes and alcoves (D-082)	A step is cut where a doorway photo stands beyond a side, or wall is seen behind it	Rooms are not all rectangles	k38 living room IoU 0.901 → 0.917, area −5.9% → −2.9%; the k65 lobby is never seen (limit)

Method	How it works	Why it helps	What it changes
Pillars and wall steps (D-083), pillar-face side (D-086)	Per photo, a run of wall 0.15–0.8 m wide standing 0.06–0.6 m proud is a pillar or step; a box side resting on a pillar face moves back to the wall	Pillars are drawn; a side on a pillar face made the room short	D-086: own lit 0 → 2 of 6 walls within 8%, own dim 4 → 5 (W3 30.0% → 1.8%); k65 and k38 unchanged
Door-anchored stitching (D-081, D-088)	The photo that looks out through a door matches the neighbour room; the two rooms snap together at that door	Turns separate rooms into one property plan with correct neighbours	k65 IoU 0.376 → 0.411, footprint –5.0% → –3.8%; fresh k65: adjacency 4/4, no overlaps, one room still drawn about 9 m off

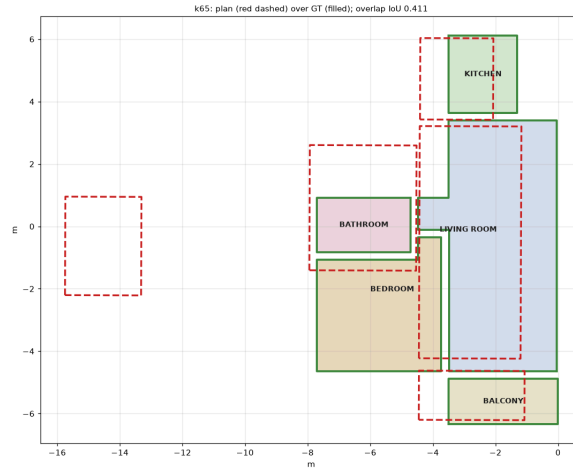


Figure 2. k65 from 31 photos (live), plan dashed over the truth: 5/5 rooms, adjacency 4/4, footprint –3.8%. One room is drawn about 9 m from its place (left) and the bedroom about 2 m off (IoU 0.41): the room graph is right, the drawing is not.

3.4 Openings

Method	How it works	Why it helps	What it changes
Doors and windows from the segmenter	Door and window pixels, cast onto the measured wall, give the opening's ends	Openings need widths, not just positions	Widths repeat within 0.2 cm between two own takes, with a +6 cm bias (BENCHMARK_REPORT.md §4)
Scoring by position (D-085)	Predicted openings pair with true ones on the same wall, nearest centre (one Hungarian match)	Pairing by width matched the wrong doors and hid errors	Two k65 "errors" of +18.0 and +35.2 cm were mismatches
Door priors (rule d, D-085)	Where a doorway is crossed but no leaf is seen, a door of standard width is placed and flagged as a prior	A walk-through stands in doorways and never sees the leaf	Crossed doorways get a door; its width is a flagged prior, not a measurement
See-through doorways	A span where depth runs past a measured wall, with both jambs in one photo, is an open doorway	Open doors are seen as holes, not as door leaves	Own bedroom door D1 found from the dim photos: 0.760 m vs 0.787 m tape; lit misses it (no photo sees it whole)

3.5 Video tier

Method	How it works	Why it helps	What it changes
DPVO poses + MoGe-2 depth	DPVO tracks the camera path up to scale; MoGe-2 gives metric depth per keyframe	No LiDAR, no poses from the phone	Window repeats to 0.4 cm SD over 9 runs; the path is the weak part
Scale from depth agreement	Keyframe pairs vote on DPVO's scale where its depth and MoGe-2's agree	Turns DPVO's path into metres	On take1 only 25–31 of 902 pairs vote, so a 13× scale drop went unseen (FIX_L00P.md)
PnP scale votes (D-075)	PnP on MoGe-2 depth between nearby keyframes gives the metric step	Many more votes, so the scale follows DPVO's drift	Same DPVO run: kf1–kf42 10.47 → 0.68 m (0.50 m true), footprint 37.8 → 29.2 m ² ; worse on blurred sample videos (D-076)
Scale step "auto" (D-087)	Per segment: PnP where its votes cover ≥ 50% of keyframes, depth agreement elsewhere	Keeps each method where it works	8 take1 caches: footprint error 0.4–66.8% → 0.6–8.7%; walls within 3% unchanged (0–3 of 14)
TSDF fusion	Scaled depth maps fused along the path into a TSDF plus a raw-point surface with a 5 cm noise σ	Same scene format as LiDAR, so the shared back end runs	1 room → 3 rooms on single_room (plan_beta v4)
SfM path gate (tried, kept off)	A global SfM path (pycolmap) used when one model holds ≥ 80% of the walk	Walk order would not matter	Replays: 2 of 3 rules fail, samples unchanged → an option only (D-087)

Method	How it works	Why it helps	What it changes
Room split by camera stays (tried, kept off)	Rooms split between places the camera stayed ≥ 4 s	Hall, passage and kitchen are one basin in video	Mixed: one cache 1 $\rightarrow$ 4 walls within 10%, another footprint error 34.5% $\rightarrow$ 43.9% (D-079)

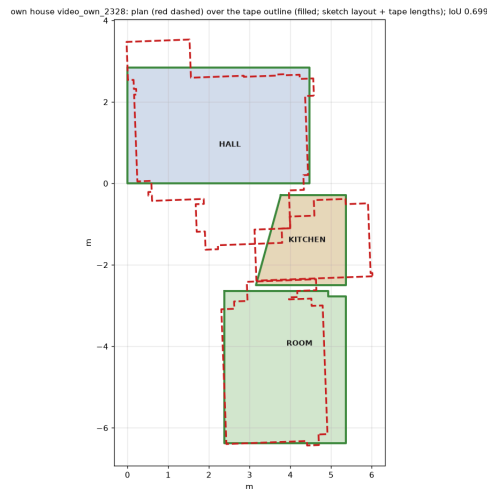


Figure 3. Own flat, video (cache own_2328): all three rooms, but overlapping; 1 of 14 walls within 3%.

3.6 Tried and dropped

What	Why dropped
Wall-first plan extractor (alpha)	Invented 5–9 m ² of rooms; four of its parts were ported (D-010, D-018)
Learned polygon models (RoomFormer, Raster2Seq)	Decimetre accuracy, research licences (D-004)
6-DoF pose graph	Tilted fragments by about 1° (modules/drift.md)
Multi-phase extractor vote	Predicted about 17/87 walls in gate, measured 11/94 (D-030)
Reference object (A4 sheet)	Made video scale worse (–18.7% vs –1.7%); opt-in only (D-067)
Door-height scale cue	Built and measured, kept off (D-069)
0.5x ultra-wide photos	Reverted to 1x (D-065, D-068)
Video frames as photos	One frame measures well, but the frames still hang on one broken path (D-080)
cuDNN deterministic flag for DPVO	DPVO still varies from frame 1 (I-007)

4. Error budget (LiDAR, one interior distance, 95%)

Term	Size	In the interval?
Sensor noise per point	σ 6–7 mm below 2 m, 16–19 mm at 2.5–4 m (D-011)	yes: fit weights and a sensor term
Plane-fit error	0.04–0.08 mm	yes, negligible
Surface bias, 0.68 cm into the room	–1.37 cm per distance before correction	corrected (D-021, D-027); residual ± 1.6 cm, ± 2.1 cm across devices
Residual drift	2.0 cm between passes, 1.0 cm within one	yes (modules/drift.md §5.5)
Topology: which surface became the wall	tens of cm when it happens	no : handled by flags and status, not σ

Camera tiers add a scale term $\sigma_{s \cdot L}$ ($2\sigma_{s \cdot A}$ for areas) and a 5 cm surface σ ; photo box sides add half their distance.

5. Calibration: do the 95% intervals hold the truth 95% of the time?

Tier, capture	Coverage	Read
Photo, own dim / lit	1.00 / 1.00 (5/5 walls each)	honest, a little wide
Photo, k65 stitched	0.97 (walls 18/18)	honest
LiDAR, k65	0.70 (walls 15/25)	too narrow: the misses are topology, not noise
LiDAR vs laser (cited)	0.93 (n 27)	close to nominal
Video, own take1	0.32–0.76 per run	too narrow: the path breaks, which no σ covers

Where the error is measurement, intervals are calibrated. Where it is topology (a missing room, a wall taken from the wrong surface) no σ covers it; the plan flags it instead (inferred walls, coverage_warning, whole_scene_consistent, reliability).

6. The fix loop (FIX_LOOP.md)

- **Declared gate:** video walls within $\pm 3\%$ on the own walkthrough: **0 of 14** in both runs (worst under a ranking rule committed beforehand).
- **Root cause:** the scale step could not follow DPVO's scale. DPVO's scale dropped about $13\times$ at $t = 34.5$ s, where the phone turned past a blank pillar; only 25-31 of 902 keyframe pairs voted, and in an earlier run of the same video none voted between $t = 12$ and 37 s.
- **Fix:** scale votes from PnP on MoGe-2 depth (D-075). **Prediction:** 0 of 14 (range 0-2), gate still fails.
- **What happened:** 3 and 0 of 14; but the old and new scale step on the same DPVO runs gave 1/1 and 1/0, so the gate did not move beyond run-to-run noise, and the plans (2 and 1 rooms against 3) missed the prediction.
- **Post-mortem:** the prediction came from replays of one single-segment run; the self-check limit, set for the old clamp, then dropped most of the walk (20 of 121 keyframes kept on one run). The fix did hold the walk together (kf1-kf42 10.47 $\rightarrow$ 0.68 m), which led to D-076 and the per-segment "auto" step (D-087).
- **Regenerate:** tags before-fix and after-fix; git diff before-fix after-fix -- floorplan/ tests/ (fixloop.diff).

7. Known failure modes (FAILURE_MODES.md)

Condition	What happens	What the pipeline does
Mirror	depth sees a reflected room; a mirror can look like a doorway	confidence filter; space not enclosed by measured walls dropped; a see-through doorway needs both jambs
Glass	no LiDAR returns, rooms merge	glazing found by the sill test; marked not_observed, never free space
Wet-look floor	phantom points below the floor ($\leq 0.7\%$)	robust floor peak; points excluded
Low light	LiDAR unaffected; camera tiers lose matches	dim own photos still gave 5/6 walls; video loses track earlier
Furniture against walls	wardrobe fronts taken as walls	wall masks, far-wall rule (D-077)
Blank walls in video	DPVO restarts or changes scale	segments, self-check, reliability = low, σ floor
Room never entered	no room	coverage_warning
Non-Manhattan wall	squared off	not handled

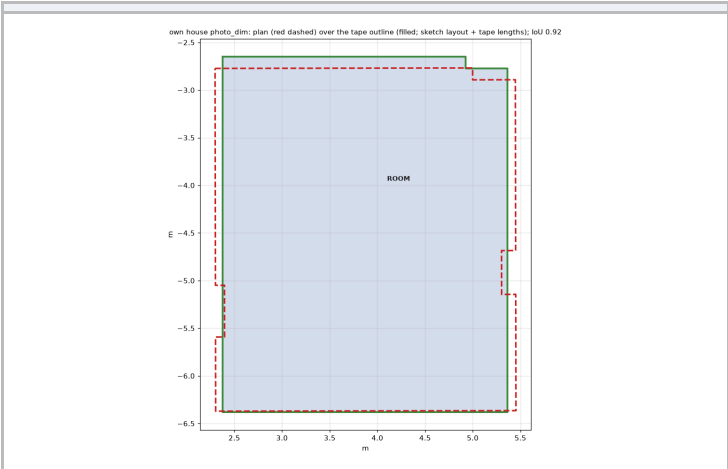


Figure 4. Own bedroom, 6 dim photos: 5/6 walls within 8%, footprint +0.6%



Figure 5. Ours vs CubiCasa 3.14.1; dimension table in BENCHMARK_REPORT.md §6

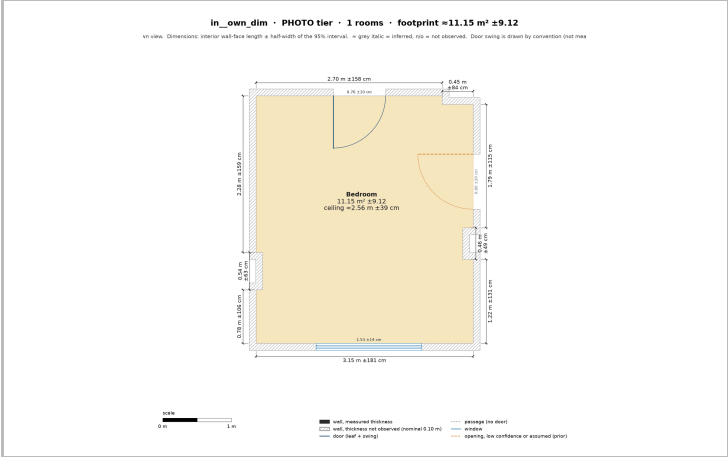


Figure 6. Dim take, technical drawing: door D1 found as a see-through doorway (0.760 m vs 0.787 m tape)	Figure 7. LiDAR, with_ceiling sample: 8 rooms, one plan, same plan on every rerun
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Deviations from the brief (no LiDAR phone, so the head-to-head uses the camera tiers; no staged damage) and all licences are in COMPLIANCE.md and DISCLOSURES.md.